

CUBIC GRAPHS OF GIRTH 11 ON 138 AND 142 VERTICES

ABSTRACT. We construct connected cubic graphs of girth 11 on 138 and 142 vertices. The order-142 graph is obtained by applying the standard two-edge subdivision construction to a recently reported order-140 graph. The order-138 graph is obtained by Biggs's excision [1, Theorem 4.2] applied to a recently reported cubic graph of order 152 and girth 12; the only points requiring verification there are that the result is connected and that its girth is 11 rather than 12. Together with the known order 140 and the previously established spectrum from order 144 onward, these constructions give $N(3, 11) \leq 138$. Neither construction is new: both operations are due to earlier authors and are applied here without modification, so the contribution is the bound.

A $(3, 11)$ -graph is a simple connected cubic graph of girth 11. Let $\text{Spec}(3, 11)$ denote the set of its possible orders, and let $N(3, 11)$ be the least even integer N such that every even $n \geq N$ belongs to $\text{Spec}(3, 11)$. Eze, Jajcay, Jajcayová, and Závacká [2] established $N(3, 11) \leq 144$ and left the even orders 128 through 142 unresolved. Exoo, Goedgebeur, Jooker, Stubbe, and Van den Eede [3] subsequently constructed three $(3, 11)$ -graphs of order 140 and a $(3, 12)$ -graph of order 152. We apply standard insertion and excision operations to two of their publicly available graphs. Neither operation is ours: the insertion is Technique 1 of [2, Section 5.1], and the excision is [1, Theorem 4.2], which applies unconditionally to any cubic graph of girth 12, so that the order-138 case is a corollary of a 1998 theorem. What is new here is only that these two operations are carried out on the graphs of [3], that the two resulting girths are verified to be exactly 11, and the bound $N(3, 11) \leq 138$ that follows.

Theorem 1. *The orders 138 and 142 belong to $\text{Spec}(3, 11)$. Consequently,*

$$N(3, 11) \leq 138.$$

All labels below are zero-based labels in the graph6 files contained in `genLifts/data/graphs/spectra.zip` in the data repository of [3].

Proposition 2. *There is a $(3, 11)$ -graph on 142 vertices.*

Proof. Let G be the graph on the first line of `spectra/n140_k3_g11.g6`. Thus G is connected and cubic, has order 140, and has girth 11. Put

$$e_1 = \{14, 87\}, \quad e_2 = \{85, 99\}.$$

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Replace e_1 by the path $14 - x - 87$, replace e_2 by the path $85 - y - 99$, and add the edge xy , where $x = 140$ and $y = 141$. Denote the resulting graph by G_{142} . This is the standard two-edge subdivision construction of [2, Section 5.1, Technique 1]; the girth is verified directly below rather than by appealing to the edge-distance hypothesis stated there, and the distances used are computed in $G - e_1 - e_2$, whereas [2] measure them in G . Since e_1 and e_2 are disjoint, G_{142} is simple, connected, and cubic.

In $G - e_1 - e_2$, breadth-first search gives

$$\text{dist}(u, v) = 8 \quad \text{for all } u \in \{14, 87\}, \quad v \in \{85, 99\}.$$

Let C be a cycle of G_{142} . If C avoids xy , suppressing every subdivision vertex used by C produces a cycle of G . Hence $|C| \geq 11$, and $|C| \geq 12$ whenever C uses x or y . If C uses xy , the remainder of C contains a path in $G - e_1 - e_2$ between an endpoint of e_1 and an endpoint of e_2 . Therefore

$$|C| \geq 1 + 1 + 8 + 1 = 11.$$

Thus G_{142} has girth at least 11. The cycle

$$0, 3, 136, 127, 108, 70, 63, 114, 125, 138, 2, 0$$

avoids e_1 and e_2 and remains an 11-cycle in G_{142} . Therefore G_{142} has girth exactly 11. $\square$

Proposition 3. *There is a $(3, 11)$ -graph on 138 vertices.*

Proof. Let G be the graph in `spectra/n152_k3_g12.g6`. Thus G is connected and cubic, has order 152, and has girth 12. Remove

$$S = \{0, 1, 2, 3, 136, 137, 144, 145, 146, 147, 148, 149, 150, 151\}$$

and write $H = G - S$. The vertices of degree 2 in H are precisely

$$D = \{112, 113, 118, 119, 122, 123, 130, 131, 134, 135, 138, 139, 140, 141, 142, 143\},$$

and every other vertex of H has degree 3. Add the perfect matching

$$M = \{\{112, 118\}, \{113, 119\}, \{122, 130\}, \{123, 131\}, \\ \{134, 140\}, \{135, 141\}, \{138, 142\}, \{139, 143\}\}.$$

Set $G_{138} = H + M$. This is precisely Biggs's excision, and not merely an operation of that type: by [1, Theorem 4.2], if there is a cubic graph with n vertices and girth g , then there is a cubic graph with $n - \epsilon(g)$ vertices and girth $g - 1$, where $\epsilon(g) = 2^{r+1} - 2$ for $g = 4r$ or $g = 4r + 1$, so that $\epsilon(12) = 2^4 - 2 = 14$; and the excised set S above is Biggs's tree, an adjacent pair together with all vertices at distance at most 2 from it (cf. [1, Example 4.5]). Applied to the order-152 graph of [3], that theorem already yields a cubic graph of order 138 and girth at least 11; what the remainder of this proof adds is only the verification that the graph so obtained is connected and that its girth is 11 rather than 12. For every edge $\{u, v\} \in M$, one has $\text{dist}_H(u, v) = 10$. Moreover, if u and v are endpoints of two distinct edges of M , then $\text{dist}_H(u, v) \geq 5$. In particular, no edge of M already

belongs to H , so G_{138} is simple. The stated degrees imply that it is cubic, and breadth-first search shows that it is connected.

Let C be a cycle in G_{138} and put

$$k = |E(C) \cap M|.$$

If $k = 0$, then C is a cycle of G , so $|C| \geq 12$. If $k = 1$, deleting the unique matching edge from C leaves an H -path of length at least 10, and hence $|C| \geq 11$. If $k \geq 2$, deleting the k matching edges decomposes C into k paths in H , each joining endpoints of distinct matching edges. Each such path has length at least 5, so

$$|C| \geq k + 5k = 6k \geq 12.$$

Thus G_{138} has girth at least 11. Finally,

$$4, 34, 97, 128, 141, 135, 124, 69, 49, 93, 32, 4$$

is an 11-cycle in G_{138} , whose only edge in M is $\{135, 141\}$. Hence G_{138} has girth exactly 11. $\square$

Proof of Theorem 1. Propositions 2 and 3 give the two asserted orders. Eze et al. [2] established that every even order at least 144 belongs to $\text{Spec}(3, 11)$, and Exoo et al. [3] supplied order 140. Therefore every even order at least 138 belongs to $\text{Spec}(3, 11)$. $\square$

No assertion is made here about the orders 128, 130, 132, 134, and 136.

Data and verification. The parent archive is available in the repository `AGT-Kulak/smallRegGirthGraphs` at commit `c2a1a4e334cd`. At that commit the archive `genLifts/data/graphs/spectra.zip` has SHA-256 checksum

`5fea94f6985d681adf33f4c104c391ea0f0ed8f80af82b4241500e5c84c9b3ac`

A verification script accompanies this note. Everything needed to reconstruct and recheck both graphs is also stated above: the two source graphs and their locations in the archive, the edges e_1 and e_2 , the excised set S , the matching M , the relevant distances, and an 11-cycle in each of G_{142} and G_{138} .

REFERENCES

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